NP-Hardness of Partial Meta-Computational Problems

Dimitrios Tsintsilidas

October 2024

Abstract

These notes are based on the paper "NP-Hardness of Learning Programs and Partial MCSP" [Hir22] and intend to complement my presentation in the Meta-Complexity Reading Group, which takes places at the University of Warwick in Term 1 2024-25. The intended result is to show the NP-hardness of the partial MINKT problem and other partial meta-computational problems. The notes do not include the complete proofs, so for a more detailed exposition, consult the original paper.

Contents

1 Introduction **1**

2 CMMSA problem **2**

3 Secret Sharing **3**

4 NW-PRG and Algorithmic Information Extraction **3**

5 Error-Correcting Code **4**

6 NP hardness of MINKT* **5**

6.1 Completeness property of the reduction 5

6.2 Soundness property of the reduction 6

6.3 Transforming the distribution 8

1 Introduction

First we define the various notions of Kolmogorov complexity used in the notes.

Definition 1.1 (Kolmogorov complexity)**.** Let $x, y \in \{0,1\}^*$ be two strings, with x_i being the ith bit of x if $i \leq |x|$ and $x_i = \perp$ otherwise, $t \in \mathbb{N} \cup \{\infty\}$ be a time bound, and A be an oracle. Then, we define the A-oracle, t-time-bounded Kolmogorov complexity of x given y as

$$\mathsf{K}^{t,A}(x \mid y) \triangleq \min\left\{|d| \mid U^{A,d,y} \text{ outputs } x_i \text{ on input } i \in [|x|+1] \text{ in time } t\right\},$$

where $U^{A,d,y}$ is a fixed universal Turing machine with oracle access to A and random access to d and y. We can omit the superscript A if $A = \emptyset$, the superscript t if $t = \infty$, and "$\mid y$" if y is the empty string. We also define the KT-complexity as

$$\mathsf{KT}(x) \triangleq \min\left\{s + t \mid \mathsf{K}^t(x) \leq s\right\}.$$

We intend to show the NP-hardness of the partial meta-computational problems below.

Definition 1.2. (Partial meta-computational problems).

- MINKT^* is defined as the language consisting of triplets of the form $(x, 1^t, 1^s)$ such that x is a partial bit-string ($x \in \{0,1,*\}^n$ for some $n \in \mathbb{N}$) and there exists a bit-string $y \in \{0,1\}^n$ consistent with x on all bits and $\mathsf{K}^t(y) \leq s$.
- MKTP^* is defined as the language consisting of pairs of the form $(x, 1^s)$ such that x is a partial bit-string ($x \in \{0,1,*\}^n$ for some $n \in \mathbb{N}$) and there exists a bit-string $y \in \{0,1\}^n$ consistent with x on all bits and $\mathsf{KT}(y) \leq s$.
- MCSP^* is defined as the language consisting of pairs of the form $(f, 1^s)$ such that f represents a partial Boolean function ($f\colon \{0,1\}^n \to \{0,1,*\}$) and there exists a circuit C of size at most s that computes some function, which is consistent with f on all bit-outputs.
- $\mathsf{NC}^1 - \mathsf{MCSP}^*$ is the language identical to MCSP^*, except that the depth of the size-s circuit is restricted to be $c \cdot \log s$, where c is a sufficiently large universal constant.

Before we show the proof of NP-hardness, we need some components used in the proof: the CMMSA problem, which we use to initiate the reduction, the secret sharing scheme, the Nisan-Wigderson pseudo-random generator and its reconstruction property in the form of algorithmic information extraction, and a well-behaved error correcting code. Each of these components are stated in the respective sections that follow.

2 CMMSA problem

In this section, we introduce the *Collective Minimum (Weight) Monotone Satisfying Assignment (CMMSA)* problem, which generalises the NP-hard problem MMSA [ABMP98, DS04]. We will reduce this NP-complete problem to the partial meta-computational problems to show their NP-completeness.

Notation: For a monotone formula ϕ on n variables and a "weight" function $w\colon [n] \to \mathbb{N}$, we define the weight of an assignment $a\colon [n] \to \{0,1\}$ with respect to w to be $w(a) \triangleq \sum_{i=1}^{n} a(i) \cdot w(i)$. We define the weight of a subset $T \subseteq [n]$ using its characteristic vector χ_T, which means $w(T) \triangleq \sum_{i \in T} w(i)$. With $\phi(a)$, we denote the output of ϕ when the variables are assigned by a.

Definition 2.1 ([Hir22])**.** The Collective Minimum (Weight) Monotone Satisfying Assignment problem with gap $g \in \mathbb{N}$ and soundness $\epsilon > 0$ ($\mathsf{CMMSA}_{g,\epsilon}$) is defined as follows:

Input: (Φ, w, θ), where $\Phi = \{\phi_1, \ldots, \phi_m\}$ is a family of monotone formulas with input variables over the set $[n]$, $w\colon [n] \to \mathbb{N}$ is a weight function represented in unary and $\theta \in \mathbb{N}$ is a threshold parameter.

Output YES: There exists an assignment $a\colon [n] \to \{0,1\}$, such that

$$\Pr_{\phi \sim \Phi}[\phi(a) = 1] = 1 \quad \text{and} \quad w(a) \leq \theta.$$

Output NO: For every assignment $a\colon [n] \to \{0,1\}$, if $w(a) \leq g \cdot \theta$, then

$$\Pr_{\phi \sim \Phi}[\phi(a) = 1] < \epsilon.$$

The degree of the family Φ is defined by $\Delta \triangleq \max_{\phi \in \Phi} |\phi|$, where $|\phi|$ is the number of literals in the formula ϕ. The size of an instance of the problem is measured by the number n of input variables.

Theorem 2.2 ([Hir22])**.** *For the parameter function $\Delta(n) \triangleq (\log n)^{1/2}$, there exists a constant $c > 0$, such that it is NP-hard under polynomial-time many-one reductions to compute ($\mathsf{CMMSA}_{g,\epsilon}$) with gap $g = \Delta(n)^c$ and soundness $\epsilon = \Delta(n)^{-c}$ on a collection Φ with degree $\Delta(n)$ of monotone DNF formulas over n variables.*

3 Secret Sharing

Notation: With $[n]$, we denote the set $\{1, \ldots, n\}$ and for $s_1, \ldots, s_n$ and a set $T \subseteq [n]$, we denote by s_T the ordered set $(s_{i_1}, \ldots, s_{i_m})$, where $T = \{i_1 < \cdots < i_m\}$.

Definition 3.1. An access structure $\mathcal{A} \subseteq 2^{[n]}$ is a "monotone" collection of subsets of $[n]$ (if $T \in \mathcal{A}$ and $T \subseteq S$, then $S \in \mathcal{A}$). If we represent a subset T by its characteristic vector $\chi_T \in \{0,1\}^n$, then we can define a monotone function $f\colon \{0,1\}^n \to \{0,1\}$ that represents $\mathcal{A}$ by $f(\chi_T) = 1 \iff T \in \mathcal{A}$ for every $T \subseteq [n]$. If $T \in \mathcal{A}$, then it is an authorised subset with respect to the access structure $\mathcal{A}$, otherwise it is unauthorised.

Definition 3.2 ([Sha79, Bla79, Bei11])**.** A secret sharing scheme for an access structure $\mathcal{A} \subseteq 2^{[n]}$ consists of a randomised algorithm Share and a deterministic algorithm Rec. The "sharing" algorithm Share takes as input a secret bit $b \in \{0,1\}$ and outputs n shares $s_1, \ldots, s_n$ that are given to n parties respectively. The "reconstruction" algorithm Rec can be used to reconstruct the bit b, while the following two properties hold:

- **Correctness:** For every authorised set $T \in \mathcal{A}$ and for every bit $b \in \{0,1\}$, the output of $\mathsf{Share}(b)$, $(s_1, \ldots, s_n)$ satisfies $\mathsf{Rec}(T, s_T) = b$ with probability 1 over the randomness used in $\mathsf{Share}(b)$.
- **Privacy:** For every unauthorised set $T \notin \mathcal{A}$ and for every random bit b on $\{0,1\}$, the random variables b and $\mathsf{Share}(b)_T$ are statistically independent.

Lemma 3.3 ([ISN93, BL90])**.** *Let $\mathcal{A}_\phi$ be the access structure represented by a monotone formula ϕ. Then, there exists a secret sharing scheme $(\mathsf{Share}(\phi, -), \mathsf{Rec}(\phi, -))$ for $\mathcal{A}_\phi$ with the required properties and also the length $|s_i|$ of each share s_i is at most the number $|\phi|$ of literals in the formula ϕ and the reconstruction algorithm $\mathsf{Rec}(\phi, T, -)$ computes a linear function of s_T over $\mathsf{GF}(2)$.*

4 NW-PRG and Algorithmic Information Extraction

The reduction that shows the NP-hardness of the partial meta-computational problems will use pseudorandom generators (PRGs), and especially for our case the Nisan-Widgerson PRG construction will be used.

Definition 4.1 (Nisan-Wigderson Pseudorandom Generator [NW94])**.** Let $d, l, m, \rho \in \mathbb{N}$ with $m \leq 2^l$. An "NW-design", $\mathcal{S}$ consists of sets $S_1, \ldots, S_m \subseteq [d]$, such that for every $i \in [m]$,

1. $|S_i| = l$, and
2. $|S_i \cap S_j| \leq \rho$, for every $j \in [m] \backslash \{i\}$.

Considering a design $\mathcal{S} = (S_1, \ldots, S_m)$, we define the pseudorandom generator

$$\mathsf{NW}_{\mathcal{S}}\colon \{0,1\}^{2^l} \times \{0,1\}^d \to \{0,1\}^m$$

as

$$\mathsf{NW}_{\mathcal{S}}(f; z) \triangleq (f(z_{S_1}), \ldots f(z_{S_m})) \in \{0,1\}^m,$$

where we consider the string $f \in \{0,1\}^{2^l}$ as the truth table of the Boolean function $f\colon \{0,1\}^l \to \{0,1\}$ and $z_{S_i} \in \{0,1\}^l$ denotes the substring of z consisting of the bits indexed by S_i.

Proposition 4.2 ([NW94, T$^+$01])**.** *For any sufficiently large $l, m, \rho \in \mathbb{N}$ with $m \leq 2^l$ and $d = O(\mathsf{exp}(l/\rho) \cdot l^2/\rho)$, there exists an "NW-design", $\mathcal{S} = (S_1, \ldots, S_m)$, which can also be constructed in time $\mathsf{poly}(2^d, m)$.*

In the original paper [NW94], it is shown that if f cannot be computed on a $(1/2 - o(1))$-fraction of inputs by small circuits, then $\mathsf{NW}_{\mathcal{S}}(f; -)$ is a pseudorandom generator secure against small circuits. We want to state this reconstruction property using Kolmogorov complexity.

Definition 4.3 ([ISW06, FLV06])**.** For any $\delta < 1/2$, any string $x \in \{0,1\}^*$, and any oracle $A \subseteq \{0,1\}^*$, we define by K^A_δ, the minimum of $\mathsf{K}^A(y)$ over all strings $y \in \{0,1\}^{|x|}$ with Hamming distance at most $\delta|x|$ from x.

We now state the algorithmic information extraction lemma for the Nisan-Wigderson pseudorandom generator:

Theorem 4.4 (Algorithmic information extraction [Hir22])**.** *Let $\mathcal{S} = (S_1, \ldots, S_m)$ be an NW-design for parameters $d, l, m, \rho \in \mathbb{N}$. Also, let $\alpha, \epsilon^{-1} \in \mathbb{N}$ and $f_1, \ldots, f_m \colon \{0,1\}^l \to \{0,1\}$ and $D \colon \{0,1\}^\alpha \times \{0,1\}^d \times \{0,1\}^m \to \{0,1\}$ be Boolean functions. Then, there exists a subset $B \subseteq [m]$ such that for every $i \in B$,*

$$\mathsf{K}^D_{\frac{1}{2}-\frac{\epsilon}{2m'}}(f_i) \leq (m-1) \cdot 2^\rho + \alpha + O(\log(m\alpha d l'/\epsilon)),$$

and for every advice $a \in \{0,1\}^\alpha$,

$$|\Pr[D(a, X_1, \ldots, X_m) = 1] - \Pr[D(a, X'_1, \ldots, X'_m) = 1]| \leq \epsilon,$$

Here, the first term is over $Z \sim \{0,1\}^d$, where we consider that the random variables X_i are defined by $X_i \triangleq f_i(Z_{S_i})$ for all $i \in [m]$, while the second term is over $Z \sim \{0,1\}^d$ and over $X'_i \sim \{0,1\}$ for all $i \in [m] \backslash B$, where we consider that the random variable X'_i for all $i \in B$ are defined exactly as X_i. Moreover, m' is the maximum, over all advice a, of the number of $i \in [m]$ such that $D(a, -, -)$ depends on X_i, and l' is $\max_{i \in [m]} \mathsf{K}^{2^{O(l)}}(f_i)$ (time-bounded Kolmogorov complexity).

5 Error-Correcting Code

The error-correcting code below is constructed using the derandomised hardness amplification theorem by Impagliazzo and Wigderson [IW97]. For the purposes of these notes, we will only use the fact below.

Lemma 5.1 ([Hir22])**.** *Given parameters $\epsilon, \delta \in (0, 1/2)$ and $\gamma > 0$, there exist a constant $c \in \mathbb{N}$ and an error correcting code that maps a string $f \in \{0,1\}^n$ to another string $\hat{f} \in \{0,1\}^{n^c}$, such that given a string $\hat{g}$ that agrees with $\hat{f}$ on a $(1/2 + \epsilon)$-fraction of bits, we can construct a $n^\gamma \cdot \mathsf{poly}(1/\epsilon\delta)$-size program with random access to the string $\hat{g}$ that outputs a string g, which agrees with f on a $(1-\delta)$-fraction of bits. Also, the next relation holds:*

$$\mathsf{K}(f) \leq \mathsf{K}_{\frac{1}{2}-\epsilon}(\hat{f}) + n^\gamma \cdot \mathsf{poly}(1/\epsilon\delta) + n \cdot \mathsf{H}_2(\delta) + O(\log n),$$

where H_2 is the binary entropy function.

Proof. Here we assume the construction of the code and prove only the relation.

From the reconstruction property, we get that

$$\mathsf{K}_\delta(f) \leq K_{\frac{1}{2}-\epsilon}(\hat{f}) + n^\gamma \mathsf{poly}(1/\epsilon\delta).$$

If h is the string of complexity $\mathsf{K}_\delta(f)$ that agrees with f on a $(1-\delta)$-fraction of bits, then the Hamming weight of $f \oplus h$ is at most δn. Thus, we can enumerate a list of all the strings with Hamming weight at most δn and choose among them:

$$\mathsf{K}(f \oplus h) \leq \log\left(\sum_{i=0}^{\delta n} \binom{n}{i}\right) + O(\log n) \leq n \cdot \mathsf{H}_2(\delta) + O(\log n).$$

If we also consider that

$$\mathsf{K}(f) \leq \mathsf{K}_\delta(f) + \mathsf{K}(f \oplus h),$$

we can get the desired equation. □

6 NP hardness of MINKT*

Theorem 6.1 ([Hir22]). *The problems* MINKT^*, MKTP^*, MCSP^* *and* $\mathsf{NC}^1 - \mathsf{MCSP}^*$ *are* NP*-hard under randomised polynomial time many-one reductions. They are also* NP*-hard to approximate within a factor of* $(\log N)^\alpha$ *on input of length* N *for a constant* $\alpha > 0$.

Proof. The proof consists of two main steps:

1. We construct a reduction from an instance of the CMMSA problem that outputs a distribution and a size bound, such that the distribution can be computed by a program of this specific size, if and only if the initial instance is a YES CMMSA-instance.

2. Using the distribution, we construct a partial bit-string, whose reconstruction properties depend on the size bound for the distribution.

Before we state the reduction, we introduce some notation. Below we assume that the family Φ has size v and degree Δ, which means that $\Phi = \{\phi_1, \dots, \phi_v\}$, where ϕ_i is a monotone formula with $|\phi_j| \leq \Delta$ for every $j \in [v]$. For each $j \in [v]$, we denote by V_j the set $\{v_1^j < \dots < v_m^j\} \subseteq [n]$ of the variables of ϕ_j. We assume that m is the number of variables that ϕ_j depends on for every $j \in [v]$.

We start with the reduction of an instance of $\mathsf{CMMSA}_{g,\epsilon}$, (Φ, w, θ), to a distribution $\mathcal{E}$ and a size bound s that have the desired properties:

Algorithm 1 Reduction R from $\mathsf{CMMSA}_{g,\epsilon}$ to a distribution

Input: (Φ, w, θ), an instance of $\mathsf{CMMSA}_{g,\epsilon}$ of size n

Output: $(\mathcal{E}, s)$, where $\mathcal{E}$ is a distribution with $\mathsf{supp}(\mathcal{E}) \subseteq \{0,1\}^{O(\log n + \Delta^2)} \times \{0,1\}$ and $s \in \mathbb{N}$ is a size bound

The reduction R defines $\lambda \triangleq \max\{(m\Delta)^2, (vn)^2, \theta, (n\Delta w_{\max})^4\}$ and picks $f_k \sim \{0,1\}^{w(k)\lambda}$ randomly for each $k \in [n]$. Then, it applies an error-correcting code $\mathsf{Enc}\colon \{0,1\}^{w(k)\lambda} \to \{0,1\}^{(w(k)\lambda)^c}$ to all f_k and it identifies the outputs $\hat{f}_k \triangleq \mathsf{Enc}(f_k)$ with functions $\hat{f}_k\colon \{0,1\}^l \to \{0,1\}$, where $l \triangleq c \cdot \log(w_{\max}\lambda)$.
▷ *The string version of* $\hat{f}_k$ *is the truth table of its function version with the remaining values equal with* 0.

Then, it sets $s \triangleq \lambda \cdot \theta$ and defines the distribution $\mathcal{E} = \mathcal{E}(\hat{f}_1, \dots, \hat{f}_n)$ from which one samples (x, b) with the following process:

1: Choose a formula $\phi_j \sim \Phi$ randomly from the family Φ, and also pick a bit $b \sim \{0,1\}$ randomly, which will be the secret;
2: Share the secret b among m parties, which means $(s_1, \dots, s_m) \triangleq \mathsf{Share}(\phi_j, b)$ with $|s_i| = \Delta$ for all $i \in [m]$;
▷ *We assume that without loss of generality, since* $|\phi_j| \leq \Delta$.
3: Construct an "NW-design" $\mathcal{S} = (S_1, \dots, S_{m\Delta})$ with $S_i \subseteq [d]$, $|S_i| = l$ and $|S_i \cap S_j| \leq \rho \triangleq l/2c$ for all $i, j \in [m\Delta]$, where $d = O(l)$, and make the clusters $\mathcal{S}_i \triangleq (S_{(i-1)\Delta+1}, \dots, S_{i\Delta})$ for every $i \in [m]$;
▷ *This means that the two desired conditions hold:* $d = O(\mathsf{exp}(l/\rho) \cdot l^2/\rho) = O(l)$ *and* $m\Delta \leq \lambda \leq 2^l$.
4: Pick $z \sim \{0,1\}^d$ randomly and set $x \triangleq (j, z, \mathsf{NW}_{\mathcal{S}_1}(\hat{f}_{v_1^j}; z) \oplus s_1, \dots, \mathsf{NW}_{\mathcal{S}_m}(\hat{f}_{v_m^j}; z) \oplus s_m)$;
▷ *We get that* $|x| = O(\log v + d + m\Delta) = O(\log n + \Delta^2)$.
5: **Output** (x, b).

6.1 Completeness property of the reduction

We will prove that assuming there exists a satisfying assignment for the family Φ with weight at most θ, then there exists a small program M that computes b on input x, for every $(x, b) \in \mathsf{supp}(\mathcal{E})$.

We start with the satisfying assignment $a : [n] \to \{0,1\}$ for which $w(a) \leq \theta$ and $\Pr_{\phi \sim \Phi}[\phi(a) = 1] = 1$. We define the set $T \triangleq \{i \in [n] \mid a(i) = 1\}$. Then the desired program M will function as follows:

Algorithm 2 Efficient program M that simulates the distribution $\mathcal{E}$

Input: $x = (j, z, \xi_1, \ldots, \xi_m)$, where $j \in [v]$, $z \in \{0,1\}^d$ and $\xi_i \in \{0,1\}^\Delta$ for all $i \in [m]$
Hard-wired input: Descriptions of the set T, the seeds $\{f_k\}_{k\in T} \in \{\{0,1\}^{w(k)\cdot\lambda}\}_{k\in T}$ and $\{(j, V_j) \mid j \in [v]\}$
Output: b such that $(x, b) \in \mathsf{supp}(\mathcal{E})$

1: Use V_j and $\{f_k \mid k \in V_j\}$ to compute $s_i \triangleq \mathsf{NW}_{\mathcal{S}_i}(\hat{f}_{v_i^j}; z) \oplus \xi_i$ for each i such that $v_i^j \in T$;
2: Run the reconstruction algorithm and output $b \triangleq \mathsf{Rec}(\phi_j, V_j \cap T, s_{V_j\cap T})$.

For the characteristic of the set $V_j \cap T$, we have $\phi(\chi_{V_j\cap T}) = \phi(\chi_T) = \phi(a) = 1$, so the set $V_j \cap T$ is authorised with respect to the access structure $\mathcal{A}_{\phi_j}$, which means that by the correctness of the secret sharing algorithm we get the correctness of M (i.e. $\Pr_{(x,b)\sim\mathcal{E}}[M(x) = b] = 1$).

Overall, we can bound the running time of M by $O(\lambda)$ or equivalently by $O(s)$[1]. Also, M has size at most

$$|T| \cdot O(\log n) + \sum_{k\in T} |f_k| + vn + O(1) \le o(\lambda) + w(T) \cdot \lambda + o(\lambda) \le (1 + o(1))\lambda\theta = O(s)$$

where we used $|T| \le \log n \le o((\lambda)^{1/2})$ and $vn \le o(\lambda)$.

Remark 6.2. We can similarly construct a circuit C of size $O(\frac{s}{\log s})$ that computes the same as the distribution, but the proof needs an error-correcting code which is list-decodable, locally-encodable and efficiently computable with query access to the input. This is achieved in [Hir22] by using the derandomised hardness amplification theorem by Impagliazzo and Wigderson [IW97]. A theorem by Uhlig is also used [Uhl74, Uhl92].

6.2 Soundness property of the reduction

Let (Φ, w, θ) be a NO instance of $\mathsf{CMMSA}_{\epsilon_0^{-1},\epsilon_0}$. Firstly, we regard the randomised reduction R successful, when $\mathsf{K}(f_{[n]}) \ge \lambda \cdot w([n]) - \log n$, which happens by a simple counting argument with probability at least $1 - 2^{\lambda\cdot w([n]) - \log n}/2^{\lambda\cdot w([n])} = 1 - \frac{1}{n}$.

We also have that for every $T \subseteq [n]$

$$\mathsf{K}(f_{[n]}) \le \mathsf{K}(f_T) + \lambda \cdot w([n]\backslash T) + O(|T| \cdot \log n),$$

since we can reconstruct $f_{[n]}$, if we have the shortest description of f_T, all the bits for $f_{[n]\backslash T}$ and the positions of f_T in a sequence of length n. The above combined with the success condition gives us

$$\lambda \cdot w(T) \le \mathsf{K}(f_T) + O(|T| \cdot \log n).$$

We will use this relation to prove the soundness property of R. We claim that for every program of size at most $s \cdot g/2$, this program fails to compute b on input x with probability at least $\frac{1}{2} - \epsilon$ for some $\epsilon > 0$ over $(x, b) \sim \mathcal{E}$.

Let M be a program of such a size and D a Boolean function that takes as advice $a = (b, j, s_1, \ldots, s_m, V_j)$ and input $(Z, Y_1, \ldots, Y_n) \in \{0,1\}^d \times (\{0,1\}^\Delta)^n$, and outputs 1 if and only if $M(j, Z, Y_{v_1^j} \oplus s_1, \ldots, Y_{v_m^j} \oplus s_m) = b$. If the program M is successful on many instances $(x, b) \in \mathsf{supp}(\mathcal{E})$, then we can imagine D as a distinguisher against the pseudorandom generator, which is against its security. This is formalised in Theorem 4.4, which we apply to the Δn strings $(\hat{f}_1, \ldots, \hat{f}_1), \ldots, (\hat{f}_n, \ldots, \hat{f}_n)$, where each $\hat{f}_k$ is repeated Δ times. Thus, we obtain a set $B \subseteq [n]$, such that for any advice a,

$$|\Pr[D(a, Z, Y_1, \ldots, Y_n) = 1] - \Pr[D(a, Z, Y'_1, \ldots, Y'_n) = 1]| \le \epsilon_0, \tag{1}$$

where the first term is over $Z \sim \{0,1\}^d$ and we consider $Y_k = \mathsf{NW}_{\mathcal{S}_k}(\hat{f}_k; Z)$ for all $k \in [n]$[2], while the second term is over $Z \sim \{0,1\}^d$ and over $Y'_k \sim \{0,1\}^\Delta$ for all $i \in [n]\backslash B$ and we consider that all Y'_k for $k \in B$ are defined exactly as Y_k.

[1] We need the running time for the proof regarding KT-complexity, so we measure it considering that M is given oracle access to the hard-wired inputs.

[2] Here we mean for all Y_i such that D depends on. It is a small detail that does not affect the result.

Moreover, $D(a,-,-)$ depends on at most $m\Delta$ bits from $Y_1,\dots,Y_n$, so from the first relation of Theorem 4.4, we get that for every $k \in B$

$$\mathsf{K}^D_{\frac{1}{2}-\frac{\epsilon_0}{2m\Delta}}(\hat{f}_k) \le (n\Delta - 1)\cdot 2^{\rho} + |a| + O(\log(2^l n\Delta |a| d/\epsilon_0)).$$

We can use that $|a| \le O(\log v) + m\Delta + n \le o(\lambda)$, $\log(2^l n\Delta|a|d/\epsilon_0) = O(\log\lambda)$, $\rho = l/2c = \log(w_{\max}\lambda)/2$ and $(n\Delta w_{\max})^4 \le \lambda$ to get

$$\mathsf{K}^D_{\frac{1}{2}-\frac{\epsilon_0}{2m\Delta}}(\hat{f}_k) \le n\Delta\cdot 2^{\log(w_{\max}\lambda)/2} + o(\lambda) \le o(\lambda).$$

Next, we use the properties of the error-correcting code to bound f_k for every $k \in B$. Using Lemma 5.1 with parameters $\gamma = 1/2$, $\delta = 1/\log n$ and $\epsilon = \epsilon_0/2m\Delta$, we get

$$\mathsf{K}^D(f_k) \le \mathsf{K}^D_{\frac{1}{2}-\epsilon}(\hat{f}_k) + |f_k|^{1/2}\cdot \mathsf{poly}(1/\epsilon\delta) + |f_k|\cdot \mathsf{H}_2(\delta) + O(\log|f_k|).$$

Considering also $\mathsf{poly}(1/\epsilon\delta) \le \lambda^{o(1)}$ and $\mathsf{H}_2(\delta) \le o(1)$, we can finally get

$$\mathsf{K}^D(f_k) \le o(\lambda\cdot w(k)),$$

for every $k \in B$. Summing them up, we have

$$\mathsf{K}^D(f_B) \le o(\lambda\cdot w(B)).$$

By mutual information and the fact that the oracle D can be computed using M, we get

$$\mathsf{K}(f_B) - |M| \le \mathsf{K}(f_B|M) + O(1) \le \mathsf{K}^D(f_B) + O(1) \le o(\lambda\cdot w(B)).$$

Combining that with the relation derived by the success condition, we have

$$\lambda\cdot w(B) \le O(|T|\log n) + |M| + o(\lambda\cdot w(B)) \le |M| + o(\lambda\cdot w(B)) \iff$$

$$(1-o(1))\cdot\lambda\cdot w(B) \le |M| \le \frac{s\cdot g}{2} = \frac{\lambda\cdot\theta\cdot g}{2} \iff w(B) < g\cdot\theta.$$

From the properties of the CMMSA problem, this means that

$$\Pr_{\phi\sim\Phi}[\phi(\chi_B) = 1] \le \epsilon_0.$$

We can rewrite Equation (1) as

$$\left|\Pr[M(j,Z,Y_{v_1^j}\oplus s_1,\dots,Y_{v_m^j}\oplus s_m) = b] - \Pr[M(j,Z,Y'_{v_1^j}\oplus s_1,\dots,Y'_{v_m^j}\oplus s_m) = b]\right| \le \epsilon_0$$

for every $j \in [v]$, every $(s_1,\dots,s_m) \in (\{0,1\}^{\Delta})^m$ and every $b \in \{0,1\}$. We can see that the distribution of the first term if we average over $j \sim [v]$, $b \sim \{0,1\}$ and take $(s_1,\dots,s_m) = \mathsf{Share}(\phi_j, b)$ is equivalent with the distribution of the input x. When we consider the second term, the shares s_i for every $i \in [m]$ such that $v_i^j \in [n]\backslash B$ can be removed, since $Y'_{v_i^j}\oplus s_i$ is uniformly random. Thus, the input of M in the second term depends only on the shares $s_{B\cap V_j}$. That means that if $\phi_j(\chi_B) = 0$, then also $\phi_j(\chi_{B\cap V_j}) = 0$, so the set $B\cap V_j$ is not authorised with respect to the access structure $\mathcal{A}_{\phi_j}$ and the shares $s_{B\cap V_j}$ are statistically independent of the bit b, i.e.

$$\Pr[M(j,Z,Y'_{v_1^j}\oplus s_1,\dots,Y'_{v_m^j}\oplus s_m) = b] = \frac{1}{2}.$$

As a result, we get

$$\Pr_{(x,b)\in\mathcal{E}}[M(x) = b] \le \Pr_{\phi\sim\Phi}[\phi(\chi_B) = 1] + \Pr[M(j,Z,Y_{v_1^j}\oplus s_1,\dots,Y_{v_m^j}\oplus s_m) = b \mid \phi_j(\chi_B) = 0] \le \epsilon_0 + \frac{1}{2} + \epsilon_0,$$

which is the desired soundness property.

6.3 Transforming the distribution

In order to construct hard instances for the partial meta-computational problems, we start by hard instances of $\mathsf{CMMSA}_{\epsilon_0^{-1},\epsilon_0}$, where $\epsilon_0 = 1/\Delta(n)^{O(1)}$, with $\Delta(n) = (\log n)^{1/2}$. We apply the reduction R, which outputs the distribution $\mathcal{E}$ and the size parameter s. Since $\log n + \Delta^2 = 2\log n$, we have that $\mathsf{supp}(\mathcal{E}) \subseteq \{0,1\}^{\log N} \times \{0,1\}$ for $N = n^{O(1)}$.

From the distribution $\mathcal{E}$, we can construct a partial Boolean function $f\colon \{0,1\}^{\log N} \to \{0,1,*\}$, such that $f(x) = b$ if there exists b such that $(x,b) \in \mathsf{supp}(\mathcal{E})$, and $f(x) = *$ otherwise. We can also identify f with a string $\tilde{f} \in \{0,1,*\}^N$, which is its truth table.

The hard instances for MCSP^* and $\mathsf{NC}^1 - \mathsf{MCSP}^*$ consist of f and the size parameter $s_{ckt} = O(\frac{s}{\log s})$ found in the completeness property against circuits, while the hard instances for MINKT^* and MKTP^* consist of $\tilde{f}$ and, the size parameter $s_{prg} = O(s)$ or the size parameter plus the time parameter $s_{prg} + t_{prg} = O(s)$, respectively, found in the completeness property against programs. We call this reduction from $\mathsf{CMMSA}_{\epsilon_0^{-1},\epsilon_0}$ to these problems, R', and we prove its correctness.

If the initial instance (Φ, w, θ) is a YES instance of CMMSA, then by the completeness property proven above, there exists a program M of size s_{prg} that runs in time t_{prg}, such that the function computed by M is consistent with f, which implies that we have YES instances of MINKT^* and MKTP^* (similarly there is a circuit C of size s_{ckt} and with the appropriate depth).

If the initial instance (Φ, w, θ) is a NO instance of CMMSA, then by the soundness property proven above, for every program of size at most $s \cdot g/2$, where $g = \epsilon_0^{-1} = \Delta(n)^{O(1)}$, we have

$$\Pr_{(x,b)\sim\mathcal{E}}[M(x) = b] \leq \frac{1}{2} + 2\epsilon_0.$$

This means that for every $y \in \{0,1\}^n$ which is consistent with $\tilde{f}$, we have $K(y) \geq s \cdot g/2 \geq s \cdot (\log N)^\alpha$, for some $\alpha > 0$. Therefore, it is NP-hard to approximate MINKT^* and MKTP^* within a factor of $(\log N)^\alpha$.

For the circuit case, assume that there is a circuit of size s' which computes a function consistent with f. If y is its truth table, we get

$$s \cdot g/2 \leq K(y) \leq O(s' \log s'),$$

from which we can derive $s' \geq \Omega(g) \cdot \frac{s}{\log s} \geq (\log N)^{\alpha'} \frac{s}{\log s}$ for some $\alpha' > 0$. Therefore, it is NP-hard to approximate MCSP^* and $\mathsf{NC}^1 - \mathsf{MCSP}^*$ within a factor of $(\log N)^{\alpha'}$. □

References

[ABMP98] Michael Alekhnovich, Sam Buss, Shlomo Moran, and Toniann Pitassi. Minimum propositional proof length is np-hard to linearly approximate. In *International Symposium on Mathematical Foundations of Computer Science*, pages 176–184. Springer, 1998. 2

[Bei11] Amos Beimel. Secret-sharing schemes: A survey. In *International conference on coding and cryptology*, pages 11–46. Springer, 2011. 3

[BL90] Josh Benaloh and Jerry Leichter. *Generalized secret sharing and monotone functions*. Springer, 1990. 3

[Bla79] George Robert Blakley. Safeguarding cryptographic keys. In *Managing requirements knowledge, international workshop on*, pages 313–313. IEEE Computer Society, 1979. 3

[DS04] Irit Dinur and Shmuel Safra. On the hardness of approximating label-cover. *Information Processing Letters*, 89(5):247–254, 2004. 2

[FLV06] Lance Fortnow, Troy Lee, and Nikolai Vereshchagin. Kolmogorov complexity with error. In *STACS 2006: 23rd Annual Symposium on Theoretical Aspects of Computer Science, Marseille, France, February 23-25, 2006. Proceedings 23*, pages 137–148. Springer, 2006. 3

[Hir22] Shuichi Hirahara. Np-hardness of learning programs and partial mcsp. In *2022 ieee 63rd annual symposium on foundations of computer science (focs)*, pages 968–979. IEEE, 2022. 1, 2, 4, 5, 6

[ISN93] Mitsuru Ito, Akira Saito, and Takao Nishizeki. Multiple assignment scheme for sharing secret. *Journal of Cryptology*, 6(1):15–20, 1993. 3

[ISW06] Russell Impagliazzo, Ronen Shaltiel, and Avi Wigderson. Reducing the seed length in the nisan-wigderson generator. *Combinatorica*, 26(6):647–681, 2006. 3

[IW97] Russell Impagliazzo and Avi Wigderson. P= bpp if e requires exponential circuits: Derandomizing the xor lemma. In *Proceedings of the twenty-ninth annual ACM symposium on Theory of computing*, pages 220–229, 1997. 4, 6

[NW94] Noam Nisan and Avi Wigderson. Hardness vs randomness. *Journal of computer and System Sciences*, 49(2):149–167, 1994. 3

[Sha79] Adi Shamir. How to share a secret. *Communications of the ACM*, 22(11):612–613, 1979. 3

[T+01] Luca Trevisan et al. Extractors and pseudorandom generators. *Journal of the ACM*, 48(4):860–879, 2001. 3

[Uhl74] Ditmar Uhlig. On the synthesis of self-correcting schemes from functional elements with a small number of reliable elements. *Matematicheskie Zametki*, 15(6):937–944, 1974. 6

[Uhl92] Dietmar Uhlig. Networks computing boolean functions for multiple input values. In *Proceedings of the London Mathematical Society Symposium on Boolean Function Complexity*, pages 165–173, 1992. 6